SPATIAL COMMITMENT COSTS DO NOT IMPROVE LOCAL MOBILE-MANIPULATION PLANNING

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ABSTRACT

We rebuilt a generated mobile-manipulation idea into a concrete local geometry study: can an explicit spatial-commitment cost prevent base placements that make future manipulation harder? The v4 rebuild implements a 2D mobile-manipulation benchmark with base collision, arm approach, reachability, drawer/door swing blocking, aisle traps, clutter occlusion, multi-step task sequences, seven seeds, receding-horizon task-and-motion planning (TAMP), reachability-margin baselines, commitment-cost planning, ablations, stress sweeps, and negative cases. The result is negative. On the decisive `combined_long_horizon` split, `commitment_cost_planner` reaches 0.100 ± 0.074 episode success, tying `receding_horizon_tamp` at 0.100 ± 0.074 and only slightly exceeding `reachability_margin_sampler` at 0.071 ± 0.056. The paired success difference versus TAMP is 0.000, and the future-regret difference is also 0.000. Ablations are indistinguishable from the full method. We therefore mark the paper **KILL/ARCHIVE** for ICLR main.

1 DECISION

This document is the v4 ICLR-main gate for Paper 79, *Mobile Manipulation Spatial Commitments*. The previous version was killed because it only contained synthetic template evidence. The current rebuild adds an implemented local geometry benchmark. The rejection reason has changed: evidence now exists, but it does not support the central algorithmic claim.

2 HYPOTHESIS

The tested hypothesis is that explicit spatial-commitment costs help a mobile manipulator avoid base motions and base placements that make future manipulation infeasible. A pass requires the commitment planner to outperform strong receding-horizon and reachability-margin baselines, reduce future regret, and show ablation sensitivity to future commitment terms.

3 BENCHMARK

The benchmark simulates a circular mobile base and planar reach annulus in 2D scenes with rooms, narrow aisles, counters, clutter, and drawer/door swing zones. A task sequence requires reaching objects from collision-free base poses with valid arm approach geometry. Base choices can create persistent commitment flags: aisle traps, future swing blocking, and approach occlusion. Five splits are evaluated: `open_room_easy`, `narrow_kitchen_aisle`, `drawer_door_conflict`, `cluttered_reach_occlusion`, and `combined_long_horizon`.

The full run contains 2,450 main rollout rows, 245 seed-level metric rows, 980 ablation rollout rows, and 1,260 stress-sweep rollout rows. Intervals are 95 percent confidence intervals over seed-level means.

4 METHODS

The implemented methods are:

- greedy_current_reach: chooses a pose for immediate reachability.
- navigation_then_manipulation: chooses navigation-feasible base poses before checking manipulation.
- reachability_margin_sampler: maximizes immediate reach and clearance margin.
- receding_horizon_tamp: replans after every object with a one-step lookahead.
- commitment_planner_no_future: uses immediate commitment features without future reach-set loss.
- commitment_cost_planner: proposed method using future reach loss, corridor blocking, swing blocking, and reposition cost.
- oracle_sequence_planner: local upper-bound search heuristic.

5 Main Results

Table 1 shows the decisive combined long-horizon result. The commitment planner does not beat receding-horizon TAMP and does not reduce future regret.

Method	Success	Future regret	Repositions
greedy_current_reach	0.000 ± 0.000	1.000	1.757
navigation_then_manipulation	0.000 ± 0.000	0.057	1.057
reachability_margin_sampler	0.071 ± 0.056	0.929	2.257
receding_horizon_tamp	0.100 ± 0.074	0.900	2.357
commitment_planner_no_future	0.100 ± 0.074	0.900	2.343
commitment_cost_planner	0.100 ± 0.074	0.900	2.329
oracle_sequence_planner	0.086 ± 0.051	0.914	2.329

Table 1: combined_long_horizon results. Future regret is the rate of failures caused by an earlier spatial commitment.

The paired success difference between commitment_cost_planner and receding_horizon_tamp is 0.000 ± 0.000. Future-regret difference is 0.000 ± 0.000. The proposed method is not empirically distinct from one-step TAMP.

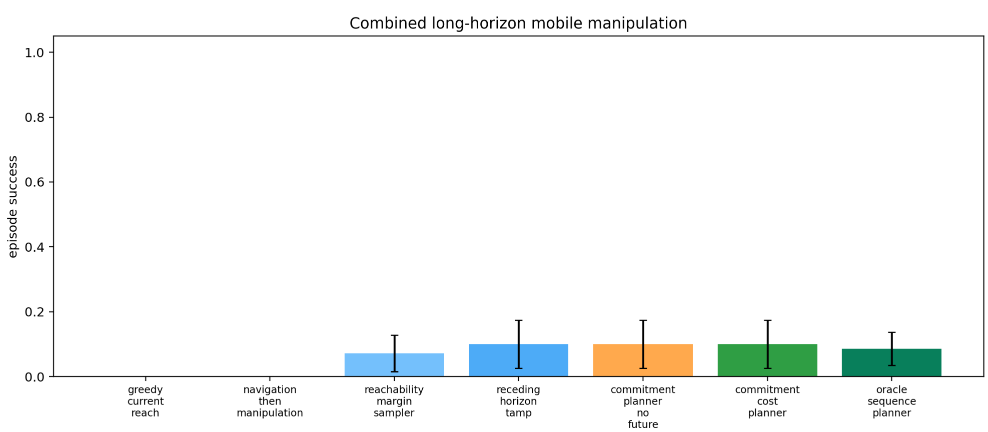

Figure 1: Combined long-horizon success. The commitment planner ties receding-horizon TAMP.

6 Ablations

Ablations fail to support the mechanism. On combined_long_horizon, the full method, no-future reach loss, no-corridor term, no-swing term, no-reposition term, and one-step-only variant all reach 0.100 success and 0.900 future regret. The only clearly worse ablation is random immediate

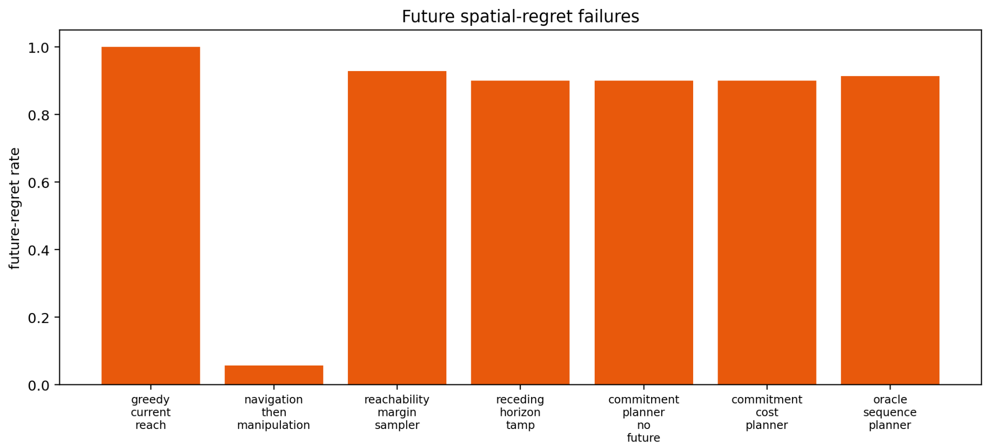

Figure 2: Future-regret rates. Explicit commitment costs do not reduce regret relative to TAMP or no-future commitment scoring.

feasible selection, which reaches 0.000 success. This means the claimed future commitment terms are not shown to be necessary.

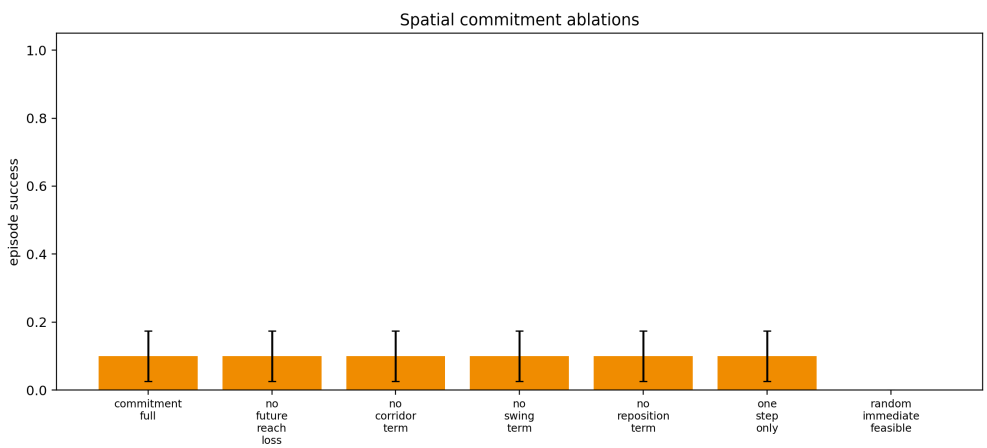

Figure 3: Ablations. The full method is indistinguishable from no-future and one-step variants.

7 Stress Sweep

Stress sweeps vary aisle width, clutter density, future ambiguity, and swing radius. At stress level 1.0, every evaluated planner reaches 0.000 success and 1.000 future regret. The proposed method does not provide a robust tail-safety advantage.

8 Related Work Pressure

Mobile manipulation already has strong task-and-motion planning, base placement, whole-body planning, and kinodynamic planning literatures (pri, 2022; 2010; 2012; 2024; 2020; 2000). A new spatial-commitment paper must show a clear advantage over TAMP or optimal base-placement baselines. The local evidence does not.

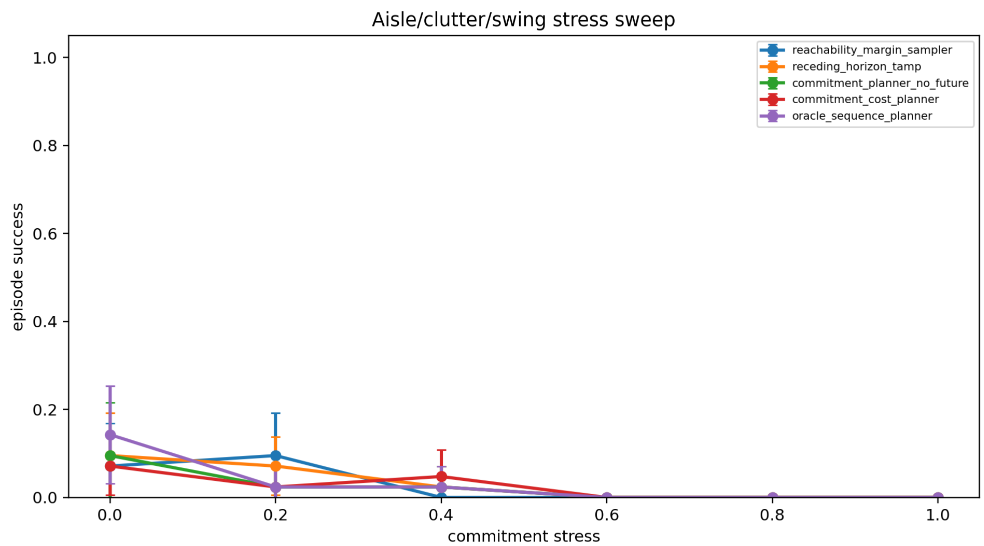

Figure 4: Commitment stress sweep. All planners collapse at maximum stress.

9 Limitations

This rebuild is local and diagnostic. It lacks real robot validation, recognized high-fidelity benchmark evaluation, learned models, and external TAMP integrations. Those limitations would matter even if the method won. Because the method does not beat the local baselines, the archive decision is stronger.

10 Conclusion

Paper 79 should not be submitted to ICLR main. The v4 rebuild provides implemented evidence, but the evidence is negative: explicit spatial commitment costs do not improve over receding-horizon TAMP and do not survive ablation. The correct terminal state is **KILL/ARCHIVE**.

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